

Homework (Habanero)

- Q1.** The slope of a line is $m = 3$. What is the slope of a line perpendicular to it?
(A) $m = -\frac{1}{3}$
(B) $m = \frac{1}{3}$
(C) $m = -3$
(D) $m = 3$
(E) $m = 0$
- Q2.** Which equation represents a line perpendicular to $y = 2x + 1$?
(A) $y = -\frac{1}{2}x + 2$
(B) $y = \frac{1}{2}x - 1$
(C) $y = 2x - 1$
(D) $y = -2x + 1$
(E) $y = x + 1$
- Q3.** In a medical trial, the dosage of a new medicine is represented by the equation $y = -\frac{3}{2}x + 12$. What is the equation of a line perpendicular to it?
(A) $y = \frac{2}{3}x + 4$
(B) $y = -\frac{2}{3}x - 4$
(C) $y = \frac{3}{2}x - 1$
(D) $y = -\frac{3}{2}x + 1$
(E) $y = \frac{1}{2}x + 2$
- Q4.** A cell is growing at a rate represented by the equation $y = 4x - 2$. What is the equation of a line that represents the rate of growth of a perpendicular cell?
(A) $y = -\frac{1}{4}x + 1$
(B) $y = \frac{1}{4}x - 1$
(C) $y = -4x + 2$
(D) $y = 4x + 2$
(E) $y = -4x - 2$
- Q5.** The equation of a line representing the relationship between the dosage of a medicine and the patient's response is $y = -2x + 5$. What is the equation of a line perpendicular to it?
(A) $y = \frac{1}{2}x + 2$
(B) $y = -\frac{1}{2}x - 2$
(C) $y = \frac{1}{2}x - 1$
(D) $y = -\frac{1}{2}x + 1$
(E) $y = 2x + 1$
- Q6.** A statistical analysis of patient data shows a relationship represented by the equation $y = \frac{3}{4}x - 1$. What is the equation of a line perpendicular to it?
(A) $y = -\frac{4}{3}x + 2$
(B) $y = \frac{4}{3}x - 2$
(C) $y = -\frac{3}{4}x + 1$
(D) $y = \frac{3}{4}x + 1$
(E) $y = -\frac{1}{4}x - 1$
- Q7.** In a medical study, the relationship between the dosage of a new vaccine and the patient's response is represented by the equation $y = -\frac{2}{5}x + 3$. What is the equation of a line perpendicular to it?
(A) $y = \frac{5}{2}x + 1$
(B) $y = -\frac{5}{2}x - 1$
(C) $y = \frac{2}{5}x - 1$
(D) $y = -\frac{2}{5}x + 1$
(E) $y = \frac{1}{5}x + 2$
- Q8.** A biologist is studying the growth of cells and finds that the rate of growth is represented by the equation $y = 5x - 2$. What is the equation of a line that represents the rate of growth of a perpendicular cell?
(A) $y = -\frac{1}{5}x + 1$
(B) $y = \frac{1}{5}x - 1$
(C) $y = -5x + 2$
(D) $y = 5x + 2$
(E) $y = -5x - 2$

Open Ended Questions (FRQ)

- Q9.** Show that the lines $y = 2x + 1$ and $y = -\frac{1}{2}x + 3$ are perpendicular. Then, find the equation of a line that is perpendicular to both of these lines.
- Q10.** A new medicine is being tested, and the relationship between the dosage and the patient's response is represented by the equation $y = -3x + 2$. Find the equation of a line that represents the rate of growth of a cell that is perpendicular to this line. Then, explain the meaning of this equation in the context of the medical study.

Chef's Tips

- Tip 1: Check that students understand the concept of perpendicular lines and their relationship to slopes.
- Tip 2: Ensure that students can apply the concept to real-world scenarios, such as medical studies.
- Tip 3: Encourage students to show their work and explain their reasoning in the free response questions.